1. LIBRARY MANAGEMENT SYSTEM USINGANGULAR AND TYPESCRIPT

GitHub link: https://github.com/MeghaSherino-2006/Library-Management-System

Team 81

Group Members List:

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4. Akhil Mohan (2460488)

Introduction:

The Library Management System is a web-based application developed using **Angular and TypeScript** to simplify and automate the management of library operations. The system enables users to browse available books, view detailed information about each book, borrow and return books, and manage member records efficiently.

This application is designed as a **Single Page Application (SPA)** using Angular's component-based architecture. It uses Angular services and dependency injection to manage data and operations. A **mock JSON server** is used to simulate backend functionality and store information related to books, members, and borrowing transactions.

The system provides a clean and interactive user interface using **Angular Material**, making it easy for users to navigate through different modules such as book listing, book details, borrowing, returning, and member management.

Objectives:

The main objectives of the Library Management System are:

- To design and develop a responsive and user-friendly library management interface.
- To allow users to view and manage book records efficiently.
- To implement borrowing and returning functionality using reactive forms.
- To apply Angular routing for smooth navigation between different pages.
- To use Angular services and dependency injection to manage application data.
- To simulate backend functionality using a Mock JSON server for CRUD operations.
- To build a modern UI using Angular Material components.

Technologies Used:

Technology	Purpose
• Angular (v17 or latest)	Frontend framework for building the application
• TypeScript	Programming language used in Angular development
• Angular Material	UI components for responsive and modern design
• HTML & CSS	Structure and styling of the application
• Node.js	Runtime environment required for Angular
• Angular CLI	Tool for creating and managing Angular projects
• JSON Server	Mock backend to simulate API calls and data storage
• Visual Studio Code	Development environment used for codin

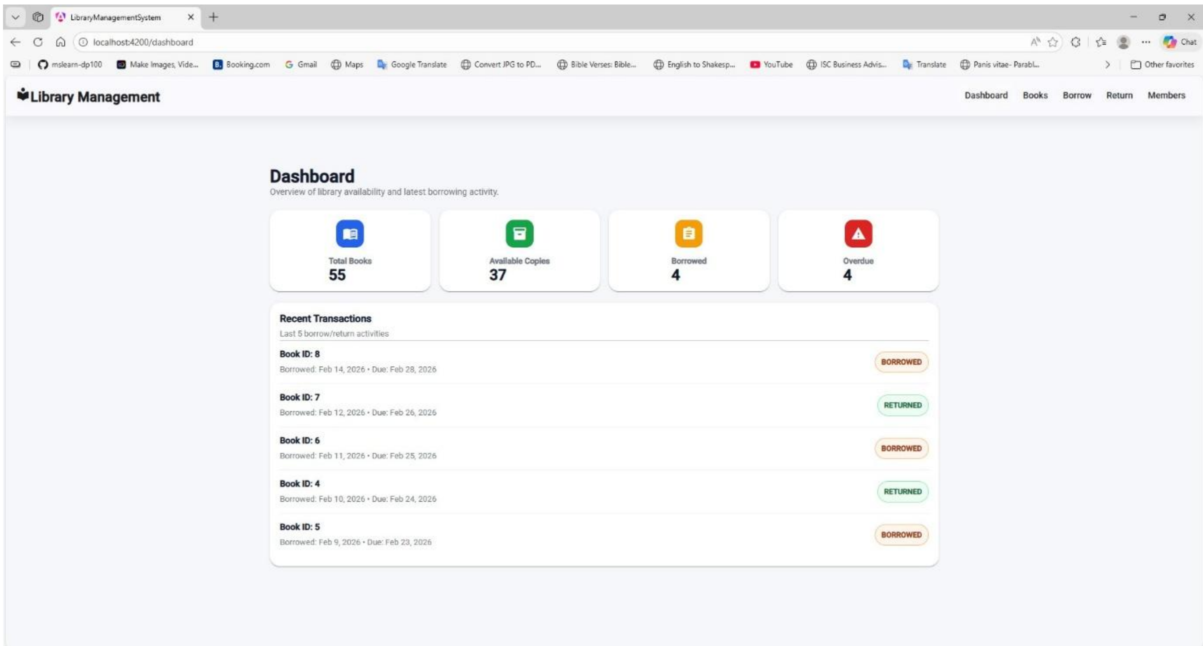
Workflow:

The workflow of the Library Management System describes how users interact with the system and how the application processes each operation.

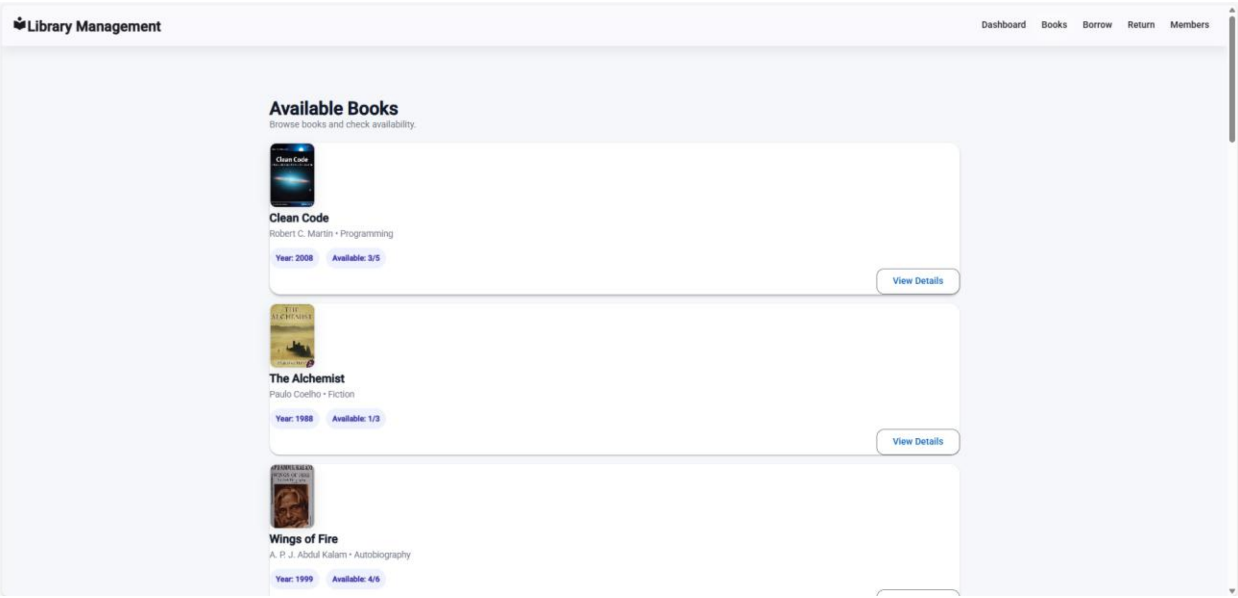
1. Application Dashboard
   The system starts with a dashboard that provides an overview of the library system.
2. Book Browsing
   Users can view a list of all available books along with details such as title, author, and availability status.
3. Book Details
   When a user selects a book, the system displays detailed information about that book.
4. Borrow Book
   Users can borrow a book through a form where the borrowing details are recorded.
5. Return Book
   Borrowed books can be returned through the return module, which updates the system records.
6. Member Management
   The system allows users to view members and register new members with proper validation.
7. Data Handling
   All book, member, and transaction data are stored and retrieved using a mock JSON server.
8. User Notifications
   Angular Material dialogs and snackbars provide confirmation messages for successful actions.

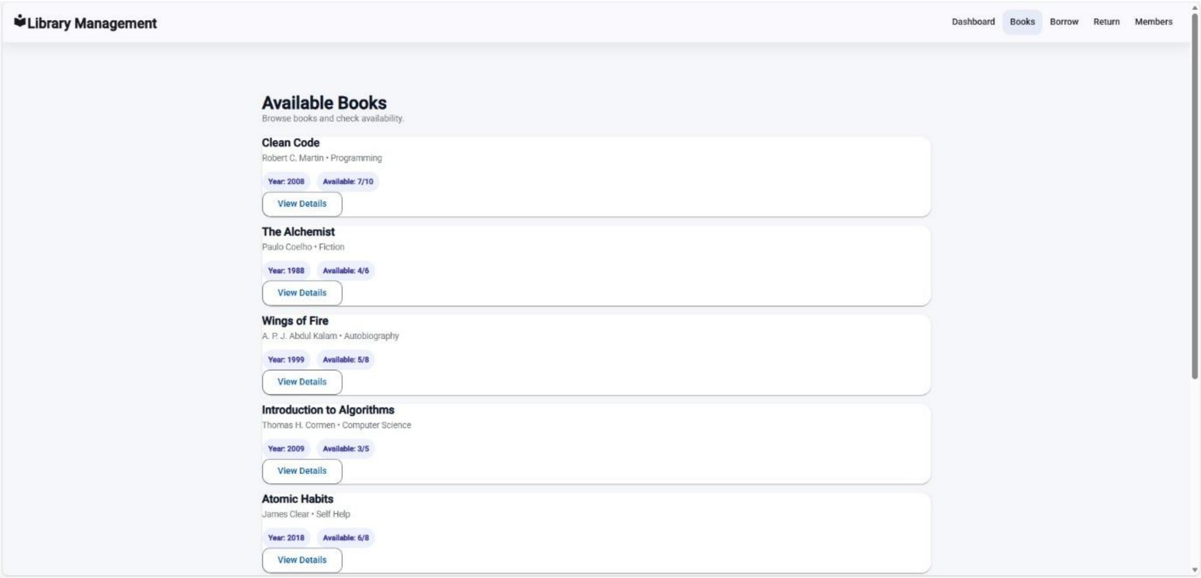
User Interface:

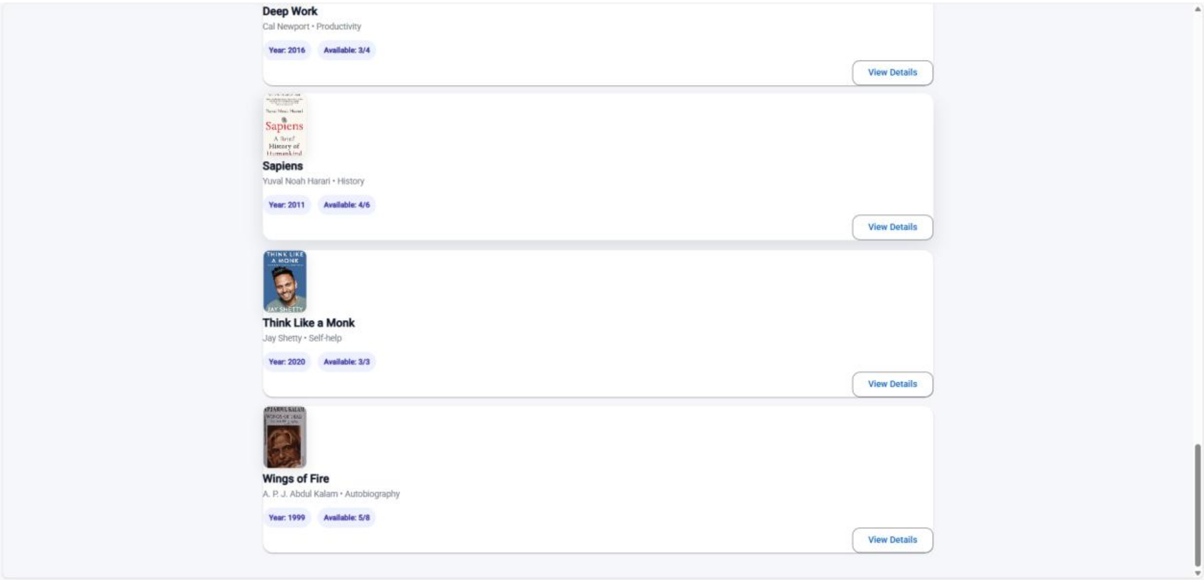
1. DASHBOARD



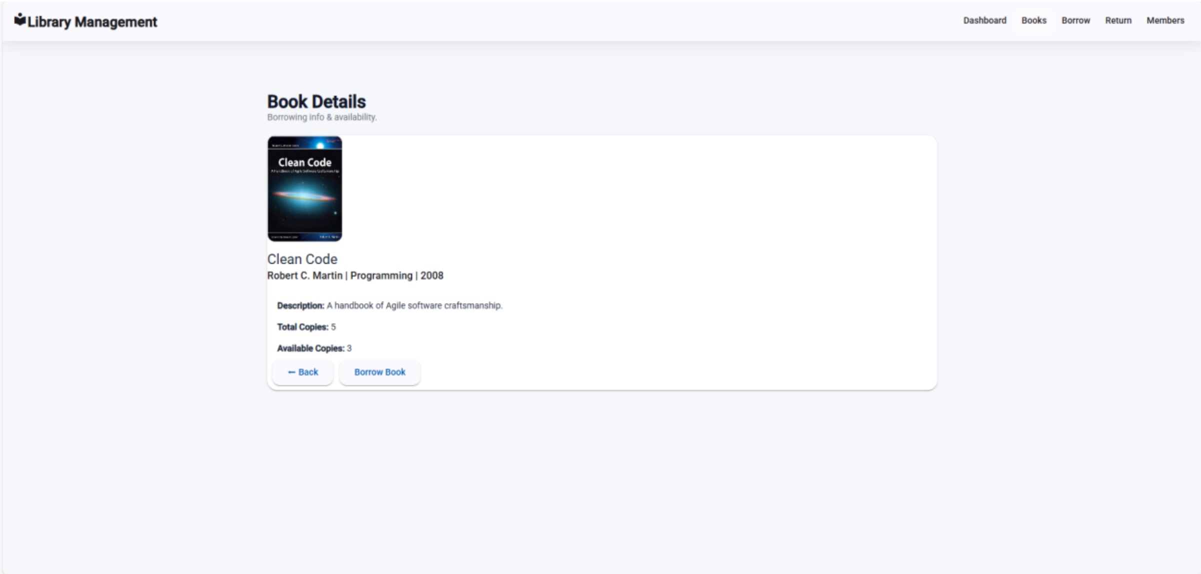
2. BOOK LIST

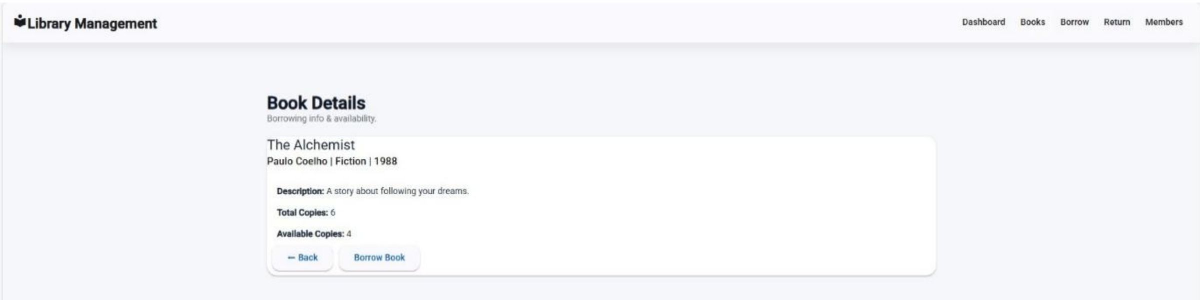



Library Management
Dashboard
Books
Borrow
Return
Members
Available Books
Browse books and check availability.
Clean Code
Robert C. Martin • Programming
Year: 2008
Available: 7/10
View Details
The Alchemist
Paulo Coelho • Fiction
Year: 1988
Available: 4/6
View Details
Wings of Fire
A. P. J. Abdul Kalam • Autobiography
Year: 1999
Available: 5/8
View Details
Introduction to Algorithms
Thomas H. Cormen • Computer Science
Year: 2009
Available: 3/5
View Details
Atomic Habits
James Clear • Self Help
Year: 2018
Available: 6/8
View Details


Deep Work
Cal Newport • Productivity
Year: 2016
Available: 3/4
View Details
Sapiens
Sapiens
Yuval Noah Harari • History
Year: 2011
Available: 4/6
View Details
Think Like a Monk
Jay Shetty • Self-help
Year: 2020
Available: 3/3
View Details
Wings of Fire
A. P. J. Abdul Kalam • Autobiography
Year: 1999
Available: 5/8
View Details

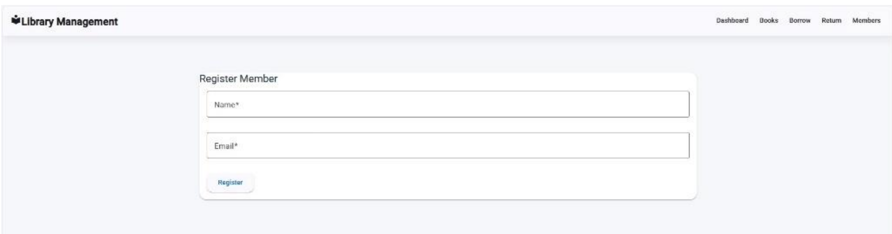
3. BOOK INFO



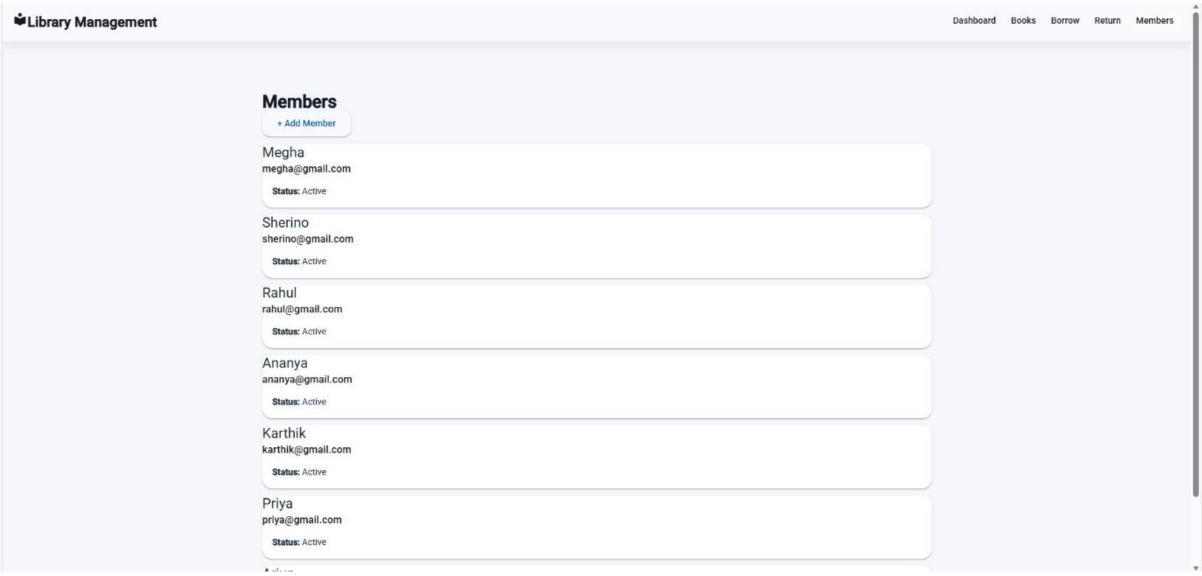


4. BORROW PAGE

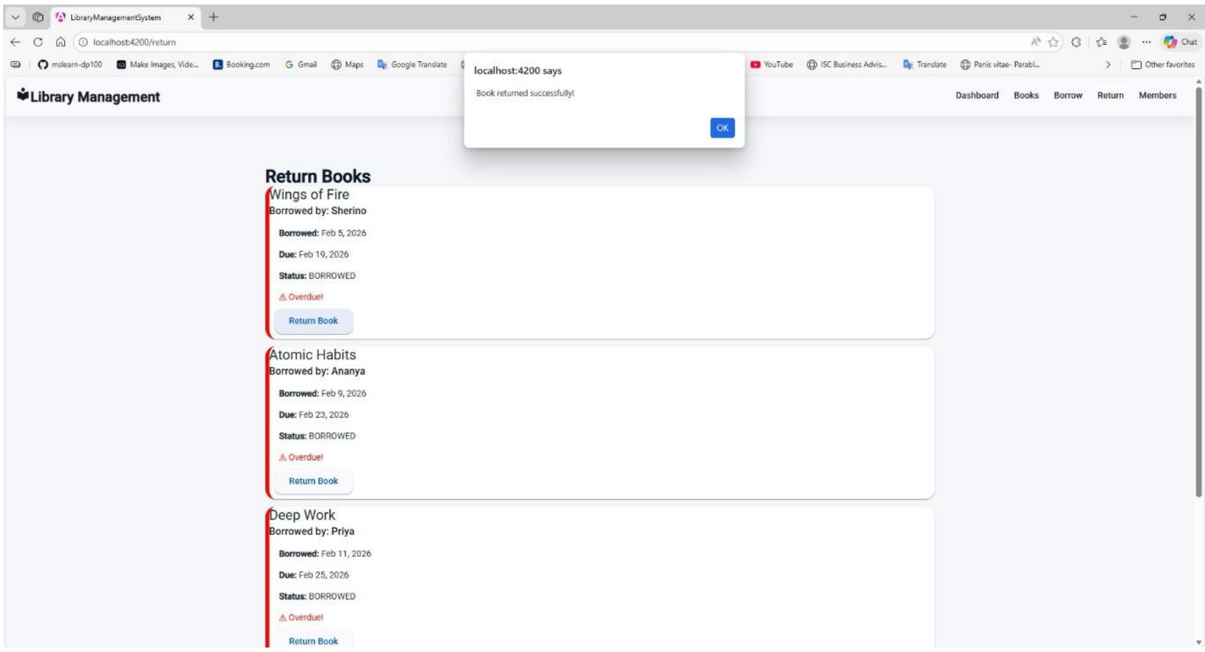
5. REGISTER MEMBER



6. MEMBERS INFO



7. RETURN BOOK



Conclusion:

The Library Management System developed using Angular and TypeScript demonstrates a modern single-page application using Angular's component-based architecture. The system allows users to browse books, view details, borrow and return books, and manage member information through a simple and interactive interface. By using Angular routing, services, reactive forms, and Angular Material, the application provides a smooth and responsive user experience while simulating backend operations using a mock JSON server.

2. Hotel Booking and Room Management System using Angular and TypeScript

GitHub link: GitHub - vibhavenu/hotel-booking-updated · GitHub

Introduction:

The Hotel Booking and Room Management System is a web-based application developed using Angular and TypeScript that allows users to browse hotels, view available rooms, and make reservations. The system also includes an admin panel that enables administrators to manage hotel rooms and booking records.

The application is built as a Single Page Application (SPA) using Angular's modular architecture. It integrates routing, reactive forms, services, and dependency injection to manage application functionality. The system uses a mock JSON server to simulate backend operations such as storing hotel information, room details, and booking records.

Angular Material components are used to create a modern and responsive user interface that improves the overall user experience.

Objectives:

The main objectives of the Hotel Booking and Room Management System are:

- To design a user-friendly hotel booking interface for browsing and reserving rooms.
- To implement hotel and room listing functionality with detailed information.
- To enable users to book hotel rooms using reactive forms.
- To provide a user dashboard where users can view their booking history.
- To implement an admin panel for managing rooms and reservations.
- To apply Angular routing for seamless navigation between pages.
- To use Angular Material UI components for a responsive design.

Technologies Used:

Technology	Purpose
Angular (v17 or latest)	Framework used for building the application
TypeScript	Programming language used in Angular development
Angular Material	UI library used for designing the interface
HTML & CSS	Used for page structure and styling
Node.js	Required runtime environment

Angular CLI	Tool for project setup and development
JSON Server	Mock backend for storing hotel, room, and booking data
Visual Studio Code	Code editor used for development

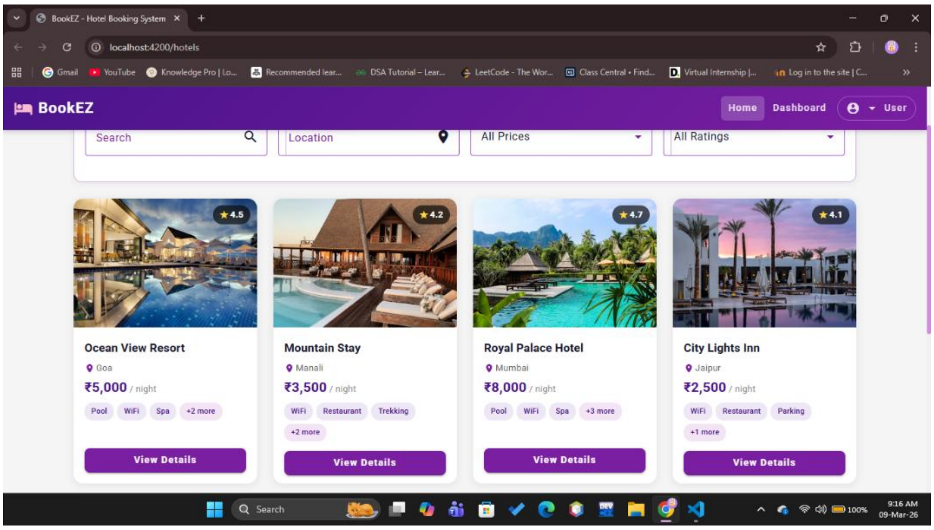
Workflow:

The workflow of the Hotel Booking System explains how users interact with the system to browse hotels and make reservations.

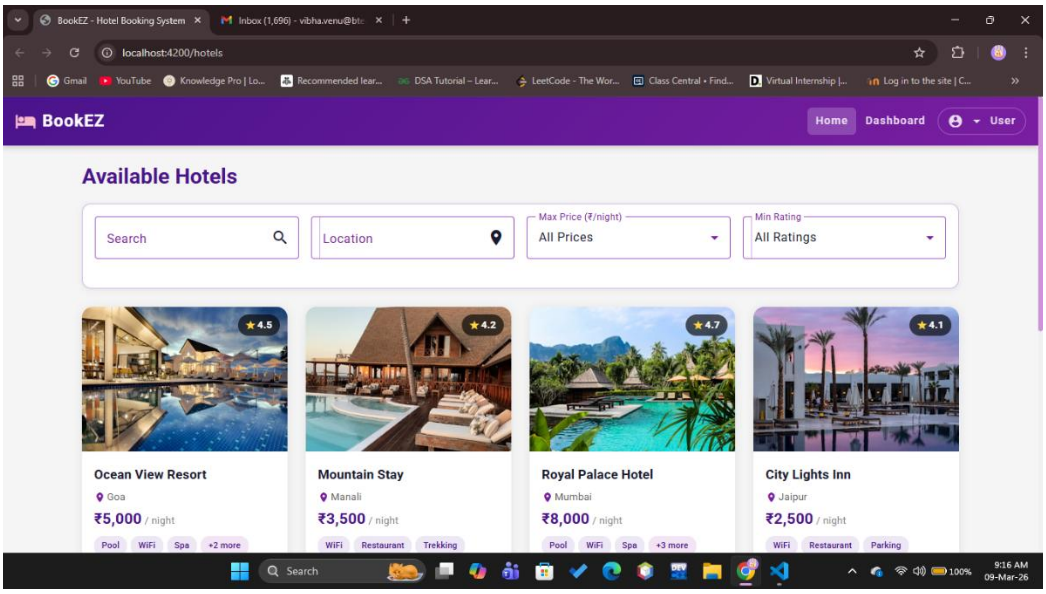
1. **User Accesses the Application**
   The user opens the hotel booking website through the browser.
2. **Hotel Listing**
   The system displays a list of available hotels along with images, location, price, and ratings.
3. **Hotel Details**
   Users can select a hotel to view detailed information such as amenities and available rooms.
4. **Room Selection**
   Users can choose a room type (standard, deluxe, or suite) depending on availability.
5. **Room Booking**
   The user fills out a booking form by entering details such as check-in date, check-out date, and number of guests.
6. **Booking Confirmation**
   After submission, the system confirms the booking and stores the booking information.
7. **User Dashboard**
   The dashboard displays the user's booking history and reservation details.
8. **Admin Management**
   Administrators can manage hotel rooms and booking records through the admin panel.

User Interface:

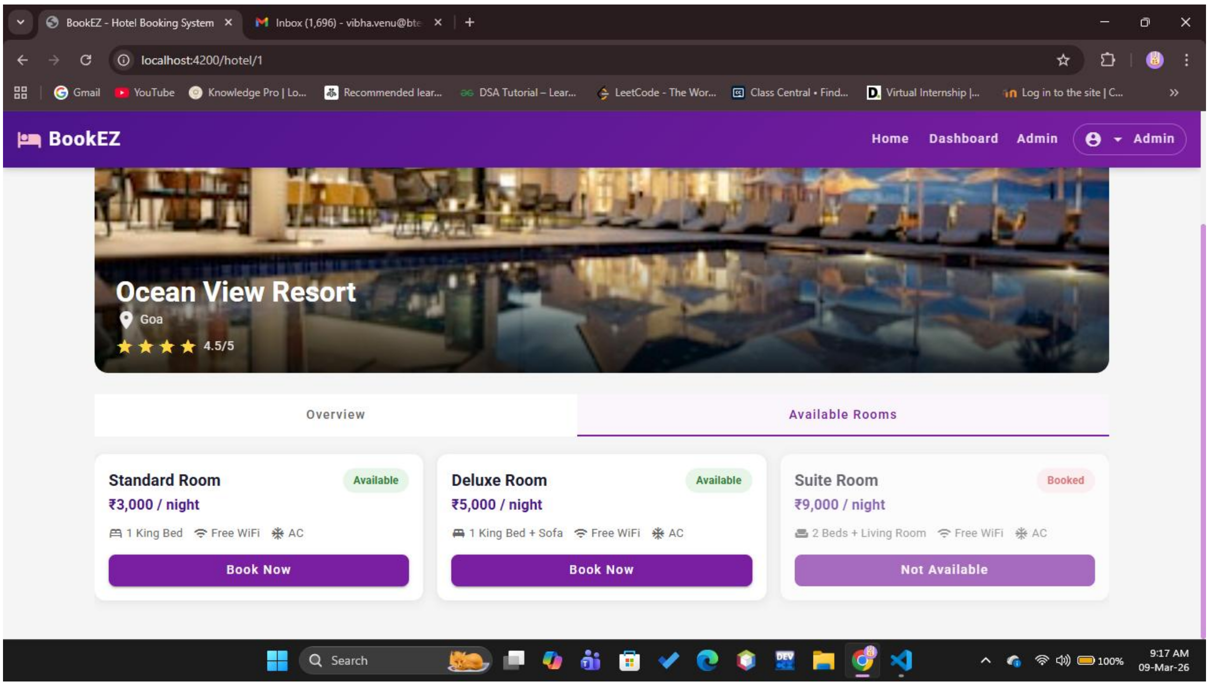
1. LANDING PAGE

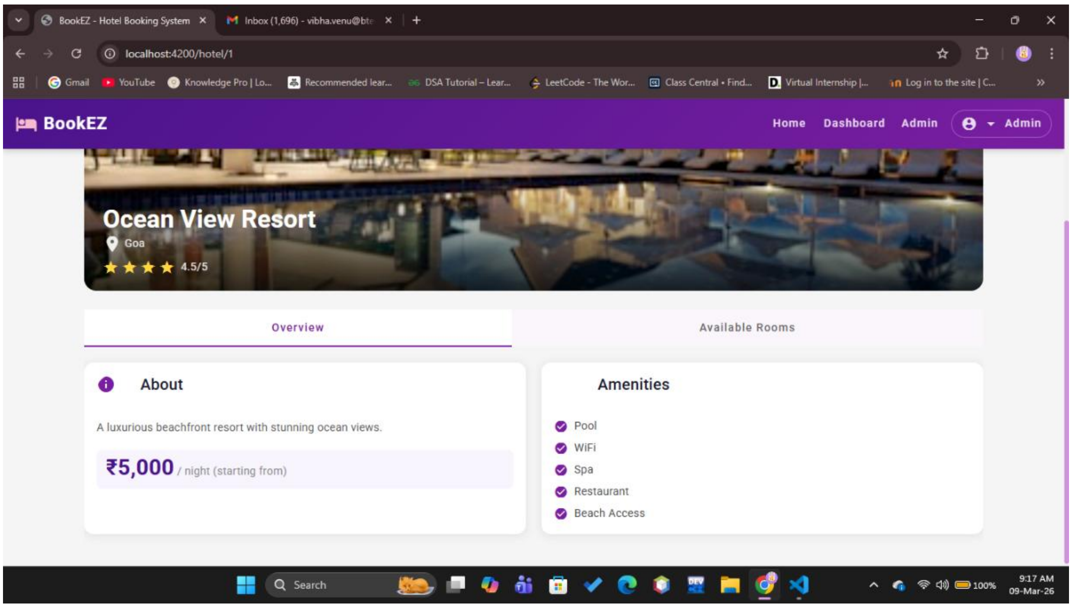


2. NAVIGATION BAR/ HOTEL LIST PAGE

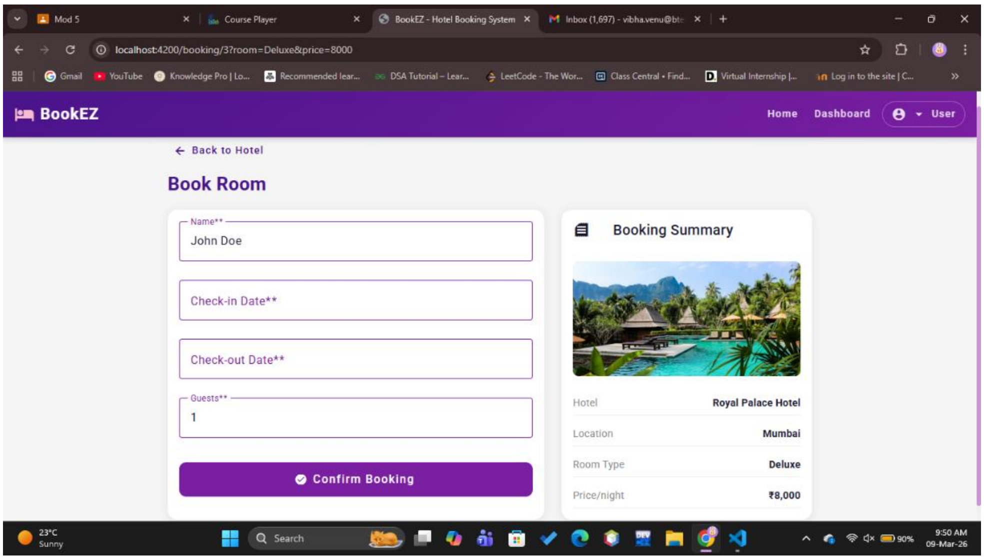


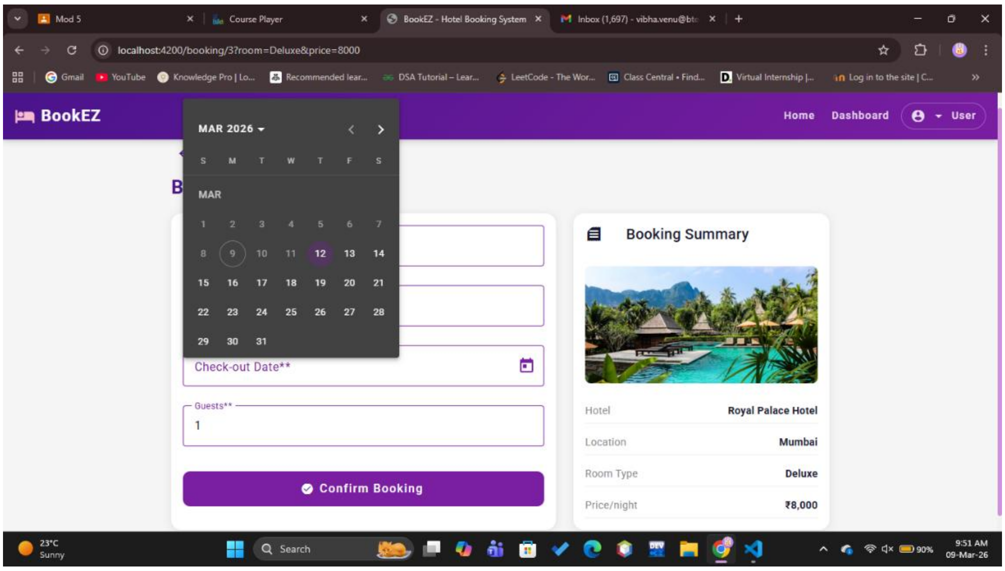
3. HOTEL DETAIL FORM



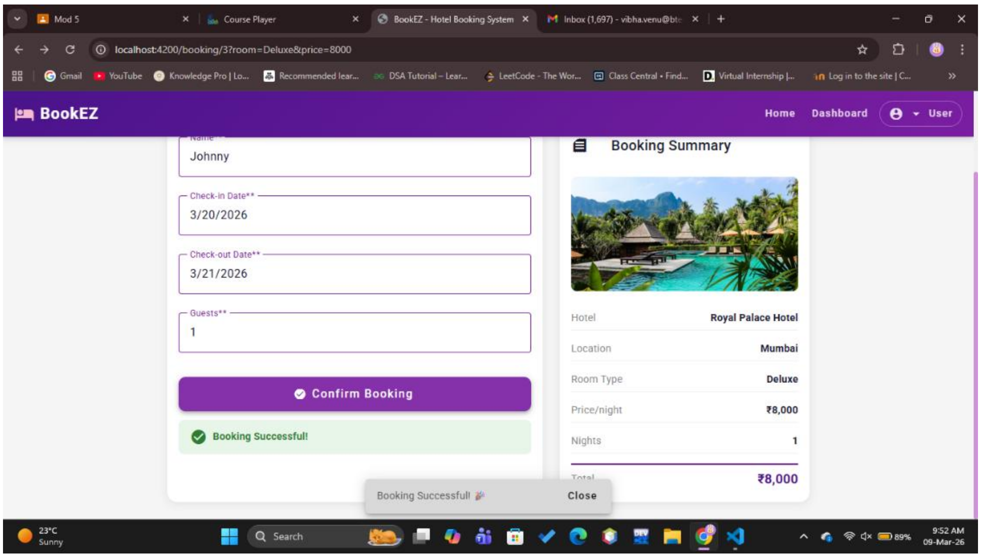


4. ROOM BOOKING FORM

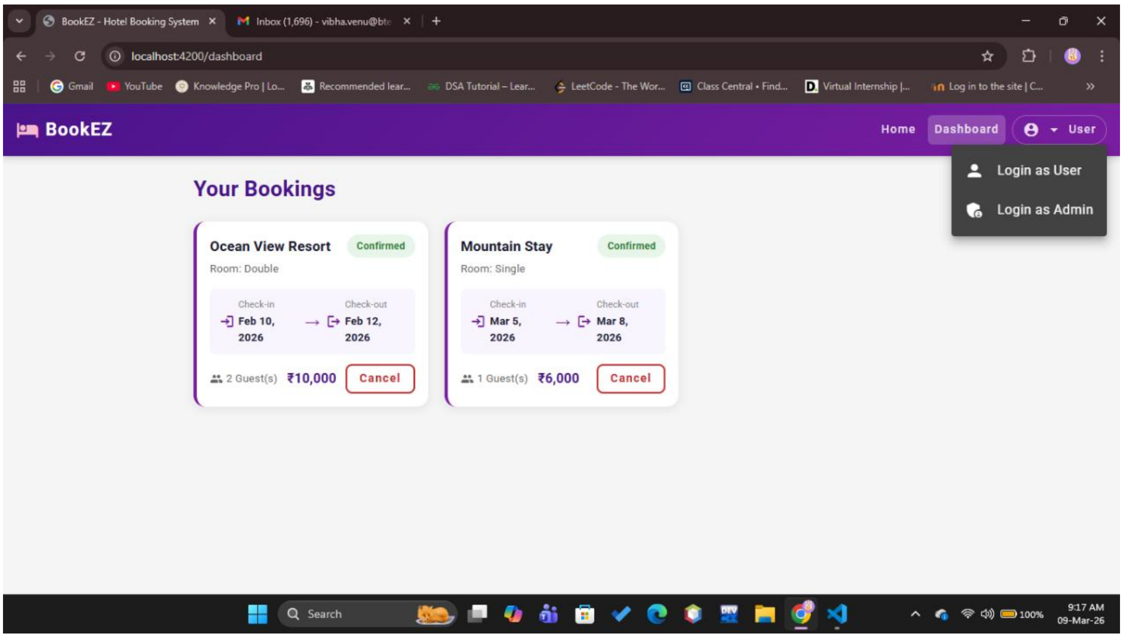




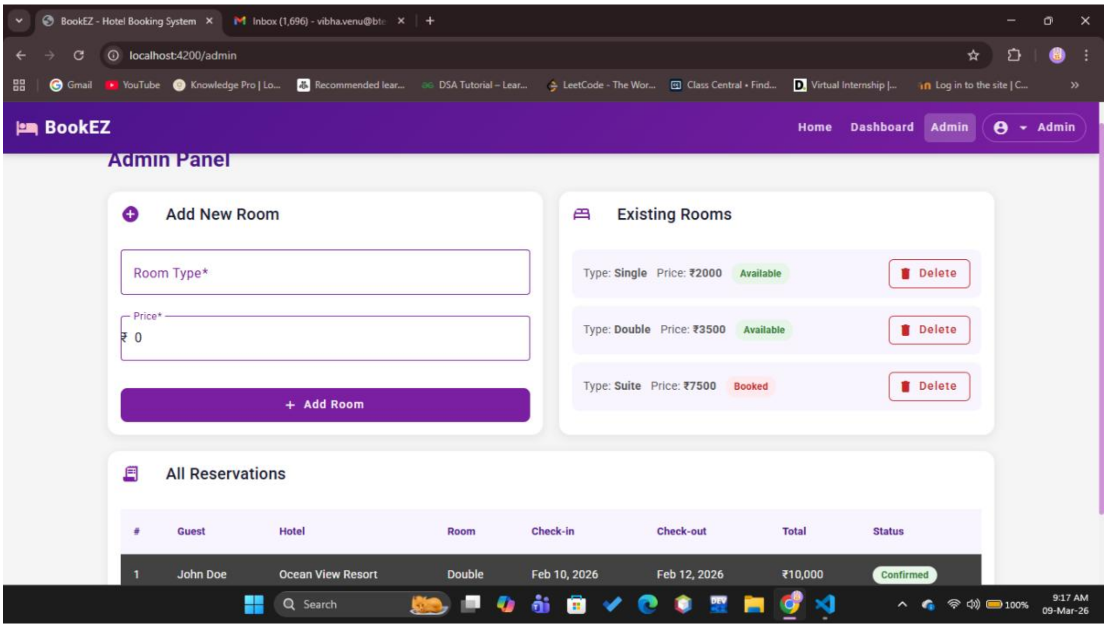
5. BOOKING CONFIRMATION DIALOG BOX



6. BOOKING CONFIRMATION/ USER DASHBOARD



7. ADMIN PANEL



Conclusion:

The Hotel Booking and Room Management System developed using Angular and TypeScript provides a comprehensive platform for users to search hotels, view available rooms, and make reservations efficiently. The system implements Angular's modular architecture, routing, reactive forms, and services to create a structured and scalable application. The use of Angular Material enhances the user interface and improves usability. The integration of a mock JSON server simulates real-world backend operations for storing hotel, room, and booking data. Through this project, important concepts such as component-based development, dependency injection, observables, and responsive UI design were applied to create a functional hotel booking application.